The self-coupled dirac16complex and its Kohn–Sham theory in gravitational fields

The nine-channel self-coupling, the 9×9 Fierz matrix, the covariant nonlinear field equation, chirality in a gravitational field, and Kohn–Sham matter in the primordial, modern and far-future epochs with its back-reaction on the expansion — sections 15–16 of the notebook 7+1.nb

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Abstract

The 16-component field dirac16complex lives on the 7+1 spacetime (one time coordinate x_0, seven space coordinates). This document builds the most general contact self-coupling of the field without derivatives — nine couplings, one for each rank of the Clifford algebra — computes the Fierz matrix that turns exchange terms into direct ones, writes the nonlinear field equation in a gravitational field with the canonical spin connection, and solves the Kohn–Sham theory of homogeneous matter of dirac16complex in the three cosmological epochs, including the self-consistent back-reaction on the expansion through the Einstein–Lovelock equations. Sections 15–16 of 7+1.nb contain 24 checks (24 True); the independent Python program tools/ref_ks_gravity.py contains 29 (29 True).

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1. Results at a glance

- **Nine channels.** With $\Theta^K = i^{\sigma_r}\Gamma^{A_1}\cdots\Gamma^{A_r}$ for the 256 increasing multi-indices K the most general self-coupling without derivatives is $\mathcal{L}_{\text{self}} = -\frac{1}{2}\sum_{r=0}^{8} g_r I_r$, $I_r = \sum_{|K|=r}(\bar\psi\Theta_K\psi)(\bar\psi\Theta^K\psi)$ (section 3).
- **Fierz.** $F_{rs} = \frac{1}{16}\sum_l (-1)^{rs-l}\binom{s}{l}\binom{8-s}{r-l}$, $F^2 = 1$; Hartree–Fock is Hartree with the couplings $G = (1-F^{\mathsf{T}})g$ (section 4).
- **Field equation.** $(i\Gamma^\mu(X)D_\mu - m)\psi = \sum_r g_r \sum_{|K|=r}(\bar\psi\Theta_K\psi)\Theta^K\psi$ with $D_\mu = \partial_\mu + \Omega_\mu$; gravity does not mix the chiral halves (section 5).

- **Kohn–Sham matter** in the cosmological spacetime: effective mass $M = m + G_S\rho_S$, vector potential $G_V\rho$; the energy density and the two pressures are integrals over the comoving Fermi ball, and $T^\mu{}_\nu$ is conserved for every expansion history (section 6).
- **Epochs.** Near $x_0 = 0$ the effective mass falls toward 0 (chiral symmetry effectively restored); in the modern epoch the matter is pressureless in ordinary space; in a de Sitter future $1 - M$ decays like the density (section 7).
- **Back-reaction.** Einstein and Einstein–Lovelock gravity driven by the Kohn–Sham matter: the constraint is preserved by the evolution to a relative 2.4×10^{-14} (section 8).

2. Conventions and objects used

Units $\hbar = c = 1$; coordinates $X = (x_0, \ldots, x_7)$ (x_0 time, x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space); frame metric $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$; repeated indices are summed. The real 16×16 Dirac matrices and their properties:

$$\Gamma_0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \quad \Gamma_i = \begin{pmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{pmatrix}, \quad \Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\eta_{AB}1_{16}, \quad \Gamma^A = \eta^{AA}\Gamma_A, \quad \bar\psi = \psi^\dagger\Gamma_0,$$

with real antisymmetric 8×8 matrices τ_i^O ($i = 1, \ldots, 7$; $(\tau_i^O)^2 = -1_8$, mutually anticommuting). The volume element and the chirality are

$$\omega = \Gamma_0\Gamma_1\cdots\Gamma_7 = \begin{pmatrix} 0 & -1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \omega^2 = -1_{16}, \qquad \chi = i\,\omega, \qquad \chi^2 = 1_{16}, \qquad P_{L,R} = \tfrac{1}{2}(1_{16} \mp \chi);$$

there is no real chirality matrix. In a gravitational field (frame $e^A{}_\mu$, inverse frame $e_A{}^\mu$, metric $g_{\mu\nu} = \eta_{AB}e^A{}_\mu e^B{}_\nu$) the generalized frame matrices, the canonical spin connection and the covariant derivative are

$$\Gamma^\mu(X) = e_A{}^\mu\Gamma^A, \qquad \omega_\mu{}^A{}_B = e^A{}_\nu\big(\partial_\mu e_B{}^\nu + \{{}^\nu_{\mu\lambda}\}e_B{}^\lambda\big), \qquad \omega_{\mu AB} = \eta_{AC}\,\omega_\mu{}^C{}_B,$$
$$\Omega_\mu = \tfrac{1}{4}\omega_{\mu AB}\Gamma^A\Gamma^B, \qquad D_\mu\psi = \partial_\mu\psi + \Omega_\mu\psi, \qquad \{{}^\lambda_{\mu\nu}\} = \tfrac{1}{2}g^{\lambda\rho}(\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}).$$

The cosmological metric used for the epochs, its Hubble rates (a prime is d/dx_0) and its spin connection:

$$ds^2 = dx_0^2 - a^2(dx_1^2 + dx_2^2 + dx_3^2) - b^2(dx_4^2 + \cdots + dx_7^2), \qquad H_a = a'/a, \quad H_b = b'/b,$$
$$\Omega_0 = 0, \quad \Omega_i = \tfrac{a'}{2}\Gamma_i\Gamma_0 \ (i = 1, 2, 3), \quad \Omega_k = \tfrac{b'}{2}\Gamma_k\Gamma_0 \ (k = 4, \ldots, 7), \quad \Gamma^\mu\Omega_\mu = \big(\tfrac{3}{2}H_a + 2H_b\big)\Gamma_0.$$

3. The nine channels of the self-coupling (section 15.1)

For a strictly increasing multi-index $K = (A_1 < \cdots < A_r)$ of rank r ($\Gamma^{()} = 1_{16}$):

$$\Theta^K = i^{\sigma_r}\,\Gamma^{A_1}\cdots\Gamma^{A_r}, \quad \sigma_r = \tfrac{r(r-1)}{2} \bmod 2, \qquad \Theta_K = \eta_K\Theta^K, \quad \eta_K = \eta_{A_1A_1}\cdots\eta_{A_rA_r}.$$

The phase makes every bilinear $\bar\psi\Theta^K\psi$ real ($\Gamma_0\Theta^K$ is Hermitian), and $\Theta_K\Theta^K = 1_{16}$, $\mathrm{tr}(\Theta_K\Theta^J) = 16\delta_{KJ}$. The invariant of rank r and the general self-coupling are

$$I_r = \sum_{|K|=r}(\bar\psi\Theta_K\psi)(\bar\psi\Theta^K\psi) = \sum_{|K|=r}\eta_K(\bar\psi\Theta^K\psi)^2, \qquad \mathcal{L}_{\mathrm{self}} = -\frac{1}{2}\sum_{r=0}^{8} g_r\,I_r.$$

The ranks have $1, 8, 28, 56, 70, 56, 28, 8, 1$ members (256 in all). The earlier Kohn–Sham theory of the notebook (section 8 of 7+1.nb) has four couplings, scalar g_S, vector g_V, axial g_A and pseudoscalar g_P:

$$\mathcal{L}_{\mathrm{self}} = -\frac{g_S}{2}(\bar\psi\psi)^2 - \frac{g_P}{2}(\bar\psi i\chi\psi)^2 - \frac{g_V}{2}(\bar\psi\Gamma^A\psi)(\bar\psi\Gamma_A\psi) - \frac{g_A}{2}(\bar\psi\Gamma^A\chi\psi)(\bar\psi\Gamma_A\chi\psi).$$

They are four of the nine:

$$g_0 = g_S, \quad g_1 = g_V, \quad g_7 = -g_A, \quad g_8 = -g_P;$$

$$I_0=(\bar\psi\psi)^2,\quad I_1=(\bar\psi\Gamma^A\psi)(\bar\psi\Gamma_A\psi),\quad I_8=-(\bar\psi\, i\chi\,\psi)^2,\quad I_7=-(\bar\psi\Gamma^A\chi\psi)(\bar\psi\Gamma_A\chi\psi).$$

$g_2,\dots,g_6$ are new (tensor channels of ranks 2 to 6). Each I_r is invariant under every local Lorentz transformation $\psi\to S\psi$, $\bar\psi\to\bar\psi S^{-1}$, $S=\exp\left(\frac18 w_{AB}[\Gamma^A,\Gamma^B]\right)$ with real antisymmetric w_{AB} ($S^{-1}=\Gamma_0 S^\dagger\Gamma_0$), so in a gravitational field it is written with the frame matrices exactly as in flat spacetime. $\Theta^K\omega$ is, up to a phase of modulus 1, Θ of the complementary multi-index (Hodge duality).

4. The Fierz matrix and Hartree–Fock with nine channels (section 15.2)

The completeness of the 256 matrices, $\sum_K(\Theta_K)_{cd}(\Theta^K)_{ab}=16\,\delta_{ad}\delta_{bc}$, rearranges every product of two bilinears. For commuting spinors a,b,c,d (anticommuting fields add a minus sign)

$$\sum_{K\in r}(a\Theta_K b)(c\Theta^K d)=\sum_{s=0}^{8}F_{rs}\sum_{J\in s}(a\Theta_J d)(c\Theta^J b),\qquad F_{rs}=\frac{1}{16}\sum_l(-1)^{rs-l}\binom{s}{l}\binom{8-s}{r-l},$$

and $F_{rs}=\frac{1}{256}\sum_{K\in r}\mathrm{tr}(\Theta_K\Theta_J\Theta^K\Theta^J)$ for any J of rank s. The matrix $16F$ (rows $r=0,\dots,8$, columns $s=0,\dots,8$):

$r\backslash s$	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
0	1	1	1	1	1	1	1	1	1
1	8	-6	4	-2	0	2	-4	6	-8
2	28	14	4	-2	-4	-2	4	14	28
3	56	-14	-4	6	0	-6	4	14	-56
4	70	0	-10	0	6	0	-10	0	70
5	56	14	-4	-6	0	6	4	-14	-56
6	28	-14	4	2	-4	2	4	-14	28
7	8	6	4	2	0	-2	-4	-6	-8
8	1	-1	1	-1	1	-1	1	-1	1

(computed here from the closed form; the matrix printed by the notebook is identical). $F^2=1$, with five eigenvalues $+1$ and four -1. With a density matrix R the direct (Hartree) energy is $\sum_s\frac{g_s}{2}D_s$, $D_s=\sum_{J\in s}\mathrm{tr}(\Theta_J R)\,\mathrm{tr}(\Theta^J R)$, and the exchange (Fock) energy is $-\sum_r\frac{g_r}{2}\sum_{K\in r}\mathrm{tr}(\Theta_K R\Theta^K R)=-\sum_{r,s}\frac{g_r}{2}F_{rs}D_s$. Hartree–Fock is therefore Hartree with the Fierz-transformed couplings

$$G=(1-F^{\mathsf T})\,g,\qquad G_s=g_s-\sum_r F_{rs}\,g_r.$$

For uniform spin-saturated matter $R=(\rho_S 1_{16}+\rho\,\Gamma_0)/16$ only $D_0=\rho_S^2$ and $D_1=\rho^2$ survive, so only G_0 and G_1 matter:

$$G_0=\tfrac{15}{16}g_0-\tfrac12 g_1-\tfrac74 g_2-\tfrac72 g_3-\tfrac{35}{8}g_4-\tfrac72 g_5-\tfrac74 g_6-\tfrac12 g_7-\tfrac{1}{16}g_8,$$
$$G_1=-\tfrac{1}{16}g_0+\tfrac{11}{8}g_1-\tfrac78 g_2+\tfrac78 g_3-\tfrac78 g_5+\tfrac78 g_6-\tfrac38 g_7+\tfrac{1}{16}g_8.$$

With the couplings of the earlier sections ($g_S=-100000$, $g_V=10000$, all others 0): $G_0=-98750$, $G_1=20000$; an additional tensor coupling $g_2=3000$ gives $(G_0,G_1)=(-104000,17375)$. With $g=(g_S,g_V,0,0,0,0,0,-g_A,-g_P)$ the exchange energy of uniform matter is that of the four-coupling theory,

$$E_x=-\frac{g_S}{32}(\rho_S^2+\rho^2)+\frac{g_P}{32}(\rho_S^2-\rho^2)-\frac{g_V}{32}(8\rho_S^2-6\rho^2)+\frac{g_A}{32}(8\rho_S^2+6\rho^2).$$

5. The covariant nonlinear field equation (section 15.3) and chirality (section 15.4)

With $\mathcal{L} = \sqrt{|g|}\,[\bar\psi(i\Gamma^\mu D_\mu - m)\psi + \mathcal{L}_{\rm self}]$ the field equation is

$$\big(i\,\Gamma^\mu(X)\,D_\mu - m\big)\psi = \sum_{r=0}^{8} g_r \sum_{|K|=r} (\bar\psi\Theta_K\psi)\,\Theta^K\psi.$$

The bilinears are Grassmann-even, so the left derivative with respect to $\bar\psi_a$ acts on them as on commuting variables; the check differentiates $\mathcal{L}_{\rm self}$ with symbolic $\bar\psi$. The chirality χ commutes with every generator $\frac14[\Gamma^A,\Gamma^B]$, hence with every Ω_μ: gravity does not mix the chiral halves. With the chiral basis

$$E_L = \frac{1}{\sqrt2}\begin{pmatrix} 1_8 \\ -i\,1_8\end{pmatrix}, \quad E_R = \frac{1}{\sqrt2}\begin{pmatrix} -i\,1_8 \\ 1_8\end{pmatrix}, \qquad \chi E_L = -E_L, \;\; \chi E_R = E_R,$$
$$\psi = E_L\phi_L + E_R\phi_R, \;\; \phi_{L,R} = E_{L,R}^\dagger\psi$$

(8-component $\phi_{L,R}$; $E_L^\dagger E_L = E_R^\dagger E_R = 1_8$, $E_L^\dagger E_R = 0$) and

$$(\tau^O\cdot\partial)_{(a,b)} = \frac1a\sum_{i=1}^{3}\tau_i^O\partial_i + \frac1b\sum_{k=4}^{7}\tau_k^O\partial_k,$$

the free Dirac equation in the cosmological spacetime becomes the two coupled 8-component equations

$$\big(i\partial_0 + i(\tfrac32 H_a + 2H_b) + (\tau^O\cdot\partial)_{(a,b)}\big)\phi_L = m\,\phi_R, \qquad \big(i\partial_0 + i(\tfrac32 H_a + 2H_b) - (\tau^O\cdot\partial)_{(a,b)}\big)\phi_R = m\,\phi_L,$$

obtained as $E_L^\dagger\Gamma_0(i\Gamma^\mu D_\mu - m)\psi$ and $E_R^\dagger\Gamma_0(i\Gamma^\mu D_\mu - m)\psi$.

6. Kohn–Sham theory in gravitational fields (section 16)

For a STATIC metric the energy of the self-coupled field is a functional of its densities (defined with the frame matrices and integrated with $\sqrt{|g|}\,d^7x$); the ground state minimizes it (Hohenberg–Kohn), and the Kohn–Sham orbitals φ_n solve the covariant Dirac equation with the mean fields,

$$\big(i\Gamma^\mu(X)D_\mu - m - \Sigma(X)\big)\varphi_n = 0, \qquad \Sigma = \sum_r g_r \sum_{|K|=r}\big[\mathrm{tr}(\Theta_K\mathcal{R})\,\Theta^K - \Theta^K\mathcal{R}\,\Theta_K\big], \qquad \mathcal{R} = \sum_{n\ \mathrm{occ}}\varphi_n\bar\varphi_n$$

(direct minus exchange; with the Fierz matrix the exchange part is again of the direct form, with G). For a time-dependent metric the same equations with the time-dependent $\mathcal{R}(X)$ are the time-dependent Kohn–Sham equations; their adiabatic limit holds when the expansion rates are small compared with the orbital energies. For homogeneous, spin-saturated matter in the cosmological spacetime the orbitals are the modes $\chi_k = e^{i(\vec p\cdot\vec x + \vec q\cdot\vec y)}c(x_0)$, $\psi = (a^3b^4)^{-1/2}\chi_k$, with comoving momenta $k = (\vec p,\vec q)$ ($\vec p$ along ordinary space $\vec x = (x_1,x_2,x_3)$, $\vec q$ along the auxiliary space $\vec y = (x_4,\dots,x_7)$) inside a comoving Fermi ball $|k| < k_F$, and the mean fields reduce to an effective mass and a vector potential:

$$i\frac{dc}{dx_0} = \Big(\sum_{i=1}^{3}\alpha_i\frac{p_i}{a} + \sum_{k=4}^{7}\alpha_k\frac{q_k}{b} + \beta M(x_0) + V_0(x_0)\Big)c, \qquad M = m + G_S\rho_S, \quad V_0 = G_V\rho,$$

with $\alpha_j = \Gamma^0\Gamma^j$, $\beta = \Gamma_0$, and $G_S \equiv G_0$, $G_V \equiv G_1$ the Hartree–Fock couplings of section 4. The physical densities are integrals over the comoving ball with $|k_a| = k\cos\theta$ (ordinary space), $|k_b| = k\sin\theta$ (auxiliary space), the solid-angle weight $\cos^2\theta\sin^3\theta$ and $\omega = \sqrt{k^2\cos^2\theta/a^2 + k^2\sin^2\theta/b^2 + M^2}$:

$$\langle f\rangle = \frac{1}{2\pi^4a^3b^4}\int_0^{k_F} dk\,k^6\int_0^{\pi/2} d\theta\,\cos^2\theta\,\sin^3\theta\,f, \qquad \rho = \langle 1\rangle = \frac{k_F^7}{105\pi^4a^3b^4}, \qquad \rho_S = \Big\langle\frac{M}{\omega}\Big\rangle,$$
$$\varepsilon = \langle\omega\rangle - \frac{(M-m)^2}{2G_S} + \frac{G_V}{2}\rho^2, \qquad p_a = \Big\langle\frac{k^2\cos^2\theta}{3a^2\omega}\Big\rangle + \Pi, \qquad p_b = \Big\langle\frac{k^2\sin^2\theta}{4b^2\omega}\Big\rangle + \Pi,$$

$$\Pi = \frac{(M-m)^2}{2G_S} + \frac{G_V}{2}\rho^2, \qquad M = m + G_S\,\rho_S(M) \;\; \text{(gap equation)}.$$

$T^\mu{}_\nu = \mathrm{diag}(\varepsilon, -p_a, -p_a, -p_a, -p_b, -p_b, -p_b, -p_b)$ is conserved for every expansion history,

$$\varepsilon' + 3H_a(\varepsilon + p_a) + 4H_b(\varepsilon + p_b) = 0,$$

because ε is stationary in M (the gap equation), which removes the term in M' (checked for $a = 1 + 0.3x_0 + 0.05x_0^2$, $b = 1/(1 + 0.2x_0)$ at $x_0 = 0.7$: residual/$\varepsilon' = -4.0 \times 10^{-12}$). The integrals are evaluated with Gauss–Legendre quadrature (96 nodes in k, 64 in θ). Units and couplings: $m = 1$; $G_S = -100000$, $G_V = 10000$. The comoving Fermi momentum is the SATURATION value $k_{F,\mathrm{sat}}$ of flat spacetime ($a = b = 1$): the Fermi momentum at which the energy per particle $E/A = \varepsilon/\rho$ is minimal, equivalently at which the pressure vanishes, $p = \rho\,\varepsilon_F - \varepsilon = 0$ (Gibbs–Duhem), with the Fermi energy $\varepsilon_F = G_V\rho + \sqrt{k_F^2 + M^2}$. In flat spacetime the notebook obtains

$$m^* = M = 0.348097340146, \qquad \rho = 1.388100213 \times 10^{-5}, \qquad \varepsilon = 1.348787233 \times 10^{-5},$$
$$\frac{E}{A} - m = \frac{\varepsilon}{\rho} - m = -0.02832142796, \qquad p_a = p_b = 0.$$

7. The three epochs (section 16.2)

Primordial — the Einstein–Kasner background $a = x_0^p$, $b = x_0^q$ with $p = (1 + 2\sqrt{2})/7$, $q = (2 - 3\sqrt{2})/14$ (normalized to $a = b = 1$ at $x_0 = 1$; it solves the vacuum equations $G^\mu{}_\nu = 0$, whose Kasner conditions are $3p + 4q = 1$, $3p^2 + 4q^2 = 1$). The density grows like $1/x_0$ ($a^3b^4 = x_0^{3p+4q} = x_0$), but the ordinary momenta are blue-shifted ($p_i/a \to \infty$), so the scalar density per particle M/ω falls and the attractive scalar coupling drives the effective mass toward zero — chiral symmetry is effectively restored near $x_0 = 0$:

x_0	10^{-4}	10^{-3}	10^{-2}	10^{-1}	1	10
m^*	0.00382418	0.0107840	0.0302385	0.0884446	0.348097	0.891764

The pressure is anisotropic: $p_a/p_b = 1.08689$, 1.64617, -0.403633 at $x_0 = 10^{-3}$, 10^{-1}, 10 (at $x_0 = 1$, flat spacetime at the saturation density, both pressures vanish).

Modern — a static auxiliary space ($b = 1$) and growing a: the ordinary momenta redshift, $M \to m$, $\varepsilon a^3 \to$ const and $p_a/\varepsilon \to 0$ — dirac16complex behaves as cold (pressureless) matter in ordinary space, while a stress remains along the auxiliary space ($p_b/\varepsilon = 0.0497735$ at $a = 100$), which the stabilizing stress of the modern epoch must balance:

a	1	2	4	10	100
m^*	0.34809734	0.85354081	0.98074107	0.99875760	0.99999876

Far future — de Sitter expansion of ordinary space $a = e^{0.1x_0}$ over $b = 1$: $\rho = \rho_0 e^{-0.3x_0}$ and $1 - M$ decays like ρ:

x_0	0	10	20	40	80
m^*	0.34809734	0.93957969	0.99692358	0.99999236	1.00000000

8. Self-consistent evolution: Einstein–Lovelock gravity driven by Kohn–Sham matter (section 16.3)

With $T^{\mu}{}_{\nu}$ of the Kohn–Sham matter the Einstein–Lovelock field equations

$$\sum_{k=1}^{3} \alpha_k E^{(k)\mu}{}_{\nu} - \bar{\Lambda}\,\delta^{\mu}_{\nu} = \kappa\, T^{\mu}{}_{\nu}, \qquad \alpha_1 = 1,$$

with the Lovelock tensors

$$E^{(k)\mu}{}_{\nu} = -\frac{1}{2^{k+1}}\,\delta^{\mu\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\nu\gamma_1\delta_1\cdots\gamma_k\delta_k}\, R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k}, \qquad \delta^{\mu_1\cdots\mu_n}_{\nu_1\cdots\nu_n} = \det\left[\delta^{\mu_i}_{\nu_j}\right],$$

($E^{(1)}$ is the Einstein tensor; $R^{\gamma\delta}{}_{\alpha\beta}$ the Riemann tensor) are a closed system for a, b. For the cosmological metric the needed components are ($H'_a = dH_a/dx_0$, $H'_b = dH_b/dx_0$):

$$\begin{aligned}
E^{(1)0}{}_{0} &= 3\left(H_a^2 + 4H_aH_b + 2H_b^2\right),\\
E^{(1)1}{}_{1} &= 2H'_a + 4H'_b + 3H_a^2 + 8H_aH_b + 10H_b^2,\\
E^{(1)4}{}_{4} &= 3H'_a + 3H'_b + 6H_a^2 + 9H_aH_b + 6H_b^2,
\end{aligned}$$

$$\begin{aligned}
E^{(2)0}{}_{0} &= -12H_b\left(4H_a^3 + 18H_a^2H_b + 12H_aH_b^2 + H_b^3\right),\\
E^{(2)1}{}_{1} &= -32H'_aH_aH_b - 48H'_aH_b^2 - 16H'_bH_a^2 - 96H'_bH_aH_b\\
&\quad - 48H'_bH_b^2 - 32H_a^3H_b - 136H_a^2H_b^2 - 192H_aH_b^3\\
&\quad - 60H_b^4,\\
E^{(2)4}{}_{4} &= -12H'_aH_a^2 - 72H'_aH_aH_b - 36H'_aH_b^2 - 36H'_bH_a^2\\
&\quad - 72H'_bH_aH_b - 12H'_bH_b^2 - 12H_a^4 - 108H_a^3H_b\\
&\quad - 180H_a^2H_b^2 - 108H_aH_b^3 - 12H_b^4,
\end{aligned}$$

$$\begin{aligned}
E^{(3)0}{}_{0} &= 360H_a^2H_b^3\left(4H_a + 3H_b\right),\\
E^{(3)1}{}_{1} &= 72H_b^2\left(8H'_aH_aH_b + 2H'_aH_b^2 + 12H'_bH_a^2 + 8H'_bH_aH_b + 8H_a^3H_b + 19H_a^2H_b^2 + 8H_aH_b^3\right),\\
E^{(3)4}{}_{4} &= 72H_aH_b\left(9H'_aH_aH_b + 6H'_aH_b^2 + 6H'_bH_a^2 + 9H'_bH_aH_b + 9H_a^3H_b + 17H_a^2H_b^2 + 9H_aH_b^3\right).
\end{aligned}$$

The 11 and 44 equations are LINEAR in (H'_a, H'_b) (Lovelock equations are quasi-linear) and are solved for them at every step; the 00 equation contains no second derivatives and is a CONSTRAINT, imposed on the initial data ($H_b(0)$ from $H_a(0)$) and preserved by the evolution because both sides are conserved. The runs ($\kappa = 1000$, $\bar{\Lambda} = 0$, $H_a(0) = 0.05$, $a(0) = b(0) = 1$, x_0 from 0 to 20; explicit Runge–Kutta of order 8 with InterpolationOrder → All, so that the solution between the steps has the order of the method) give:

gravity	$H_b(0)$	$a(20)$	$b(20)$	$H_a(20)$	$H_b(20)$	**max. relative constraint violation**
Einstein ($\alpha_2 = \alpha_3 = 0$)	0.0091437124	1.5817387	1.1975586	0.011877999	0.0074191078	2.0×10^{-14}
Einstein–Lovelock ($\alpha_2 = 20$, $\alpha_3 = 500$)	0.014980704	1.4773515	1.2760841	0.010175914	0.0084887647	2.4×10^{-14}

9. Verification

Every statement above is a check of 7+1.nb (Wolfram Language 15.0.1) and of the independent Python program tools/ref_ks_gravity.py (numpy, scipy, sympy), which rebuilds the Dirac matrices and computes everything again with its own quadrature, gap solver and Runge–Kutta integrator (DOP853). The numbers printed by both programs:

quantity	**7+1.nb**	**Python**	**max. relative difference**
flat limit: m^*, ρ, ε	0.34809734, 1.3881002×10^{-5}, 1.3487872×10^{-5}	0.34809734, 1.3881002×10^{-5}, 1.3487872×10^{-5}	7.5×10^{-15}
primordial $m^*(x_0)$	0.0038241794, 0.010783990, 0.030238520, 0.088444641, 0.34809734, 0.89176375	0.0038241794, 0.010783990, 0.030238520, 0.088444641, 0.34809734, 0.89176375	5.3×10^{-15}
primordial p_a/p_b	1.0868869, 1.6461738, -0.40363339	1.0868869, 1.6461738, -0.40363339	2.3×10^{-15}
modern $m^*(a)$	0.34809734, 0.85354081, 0.98074107, 0.99875760, 0.99999876	0.34809734, 0.85354081, 0.98074107, 0.99875760, 0.99999876	4.1×10^{-15}
modern p_b/ε at $a = 100$	0.049773512	0.049773512	3.1×10^{-15}
far future $m^*(x_0)$	0.34809734, 0.93957969, 0.99692358, 0.99999236, 1.00000000	0.34809734, 0.93957969, 0.99692358, 0.99999236, 1.00000000	4.1×10^{-15}
Einstein run: $H_b(0)$; a, b, H_a, H_b at $x_0 = 20$	0.0091437124, 1.5817387, 1.1975586, 0.011877999, 0.0074191078	0.0091437124, 1.5817387, 1.1975586, 0.011877999, 0.0074191078	2.6×10^{-12}
Einstein–Lovelock run: $H_b(0)$; a, b, H_a, H_b at $x_0 = 20$	0.014980704, 1.4773515, 1.2760841, 0.010175914, 0.0084887647	0.014980704, 1.4773515, 1.2760841, 0.010175914, 0.0084887647	2.1×10^{-12}

The largest relative difference is 2.6×10^{-12}.

The checks of sections 15–16 of the notebook:

#	**check**	**result**	**section**
1	`256 multi-indices in 9 ranks with 1, 8, 28, 56, 70, 56, 28, 8, 1 members`	True	15.1
2	`every bilinear psibar Theta^K psi is real: Gamma17[0].Theta^K Hermitian for all 256 K`	True	15.1
3	`Theta_K.Theta^K == ID16 and Tr[Theta_K.Theta^J] == 16 delta_KJ`	True	15.1
4	`Hodge duality: Theta^K.omega17 == (phase) Theta^(complement of K), |phase| = 1`	True	15.1
5	`the couplings of section 8 are four of the nine: I_0 == (psibar psi)^2, I_1 == (psibar Gamma^A psi)(psibar Gamma_A psi), I_8 == -(psibar i chi17 psi)^2, I_7 == -(axial invariant)`	True	15.1
6	`local Lorentz invariance: I_r(S psi, psibar S^-1) == I_r(psi, psibar) for r = 0..8 (random finite Lorentz transformation), and S^-1 == Gamma17[0].S^dagger.Gamma17[0]`	True	15.1
7	`Fierz matrix: closed form == (1/256) Sum_{K in r} Tr[Theta_K.Theta_J.Theta^K.Theta^J] for the first and last J of every rank s`	True	15.2
8	`F.F == IdentityMatrix[9]; eigenvalues: five +1 and four -1`	True	15.2
9	`Fierz identity for commuting spinors (random): Sum_{K in r} (a Theta_K b)(c Theta^K d) == Sum_s F_rs Sum_{J in s} (a Theta_J d)(c Theta^J b), r = 0..8`	True	15.2
10	`Hartree-Fock energy E_H + E_x == Sum_s (G_s/2) D_s with G = (1 - Transpose[F]) g (random density matrix and couplings)`	True	15.2
11	`uniform density matrix (rhoS ID16 + rho Gamma17[0])/16: D_0 == rhoS^2, D_1 == rho^2, D_2 .. D_8 == 0`	True	15.2
12	`section 8 recovered: with g = (gS, gV, 0, 0, 0, 0, 0, -gA, -gP) the exchange energy of uniform matter is that of section 8.7`	True	15.2
13	`the Hartree-Fock couplings of section 8.7: gS = -100000, gV = 10000 give G_0 = -98750, G_1 = 20000`	True	15.2
14	`nonlinear field equation: dL_self/d psibar_a == -Sum_r g_r Sum_K (psibar Theta_K psi) (Theta^K psi)_a`	True	15.3
15	`chi17 commutes with every generator (Gamma17up[A].Gamma17up[B] - Gamma17up[B].Gamma17up[A])/4, hence with every Omega_mu`	True	15.4
16	`curved chiral equations: E_L^dag.Gamma17[0].(i Gamma^mu D_mu - m) psi == (i d0 + i(3Ha/2 + 2Hb) + tauO.d) phi_L - m phi_R, and the same for E_R with -tauO.d`	True	15.4
17	`flat limit a = b = 1, kF = kFsat: m* == 0.3480973401... and E/A - m == -0.0283214279... (section 8.7), to 10^-11`	True	16.1
18	`flat limit: pa == pb == rho eps_F - eps (Gibbs-Duhem), which is 0 at the saturation density`	True	16.1

#	check	result	section
19	energy-momentum conservation of the Kohn-Sham matter: eps' + 3Ha(eps + pa) + 4Hb(eps + pb) == 0 (a = 1 + 0.3x + 0.05x^2, b = 1/(1 + 0.2x), x0 = 0.7; relative 10^-8)	True	16.1
20	primordial: rho x0 == kFsat^7/(105 Pi^4) at every x0, and m* decreases monotonically to below 0.05 at x0 = 10^-4	True	16.2
21	modern (b = 1): eps a^3 tends to a constant and pa/eps -> 0 (pressureless in ordinary space)	True	16.2
22	far future (de Sitter, H = 0.1): rho == rho0 e^{-3 H x0} and 1 - m* decays like rho	True	16.2
23	the 11 and 44 Einstein-Lovelock equations are linear in (Ha', Hb'); the 00 equation contains neither (constraint)	True	16.3
24	self-consistent evolution: the constraint Sum alpha_k E^(k)0_0 - Lambda == kappa eps is preserved to 10^-7 (relative) for Einstein and for Einstein-Lovelock (alpha2 = 20, alpha3 = 500)	True	16.3

The Python program's boolean checks:

#	check	result
1	256 products in 9 ranks: 1, 8, 28, 56, 70, 56, 28, 8, 1	True
2	every bilinear psibar Theta^K psi is real: Gamma17[0] Theta^K Hermitian (Theta^K = i^((r(r-1)/2) mod 2) Gamma^K)	True
3	Theta_K Theta^K = 1_16 and tr(Theta_K Theta^J) = 16 delta_KJ	True
4	Fierz completeness: sum_K (Theta_K)_cd (Theta^K)_ab = 16 delta_ad delta_bc	True
5	Hodge duality: Theta^K omega17 = +- Theta^(complement of K) (up to a phase)	True
6	stage-2 channels: I_0 = (psibar psi)^2, I_1 = (psibar Gamma^A psi)(psibar Gamma_A psi), I_8 = -(psibar i chi psi)^2, I_7 = -(axial)	True
7	local Lorentz invariance: I_r(S psi, psibar S^-1) = I_r(psi, psibar), r = 0..8 (random finite boost-rotation)	True
8	Fierz matrix F_rs = (1/256) sum_{K in r} tr(Theta_K Theta_J Theta^K Theta^J), independent of the representative J of rank s	True
9	Fierz matrix closed form F_rs = (1/16) sum_l (-1)^(rs-l) C(s,l) C(8-s,r-l)	True
10	F^2 = 1 (the Fierz transformation is an involution)	True
11	Fierz identity for commuting spinors: sum_{K in r} (a Theta_K b)(c Theta^K d) = sum_s F_rs sum_{J in s} (a Theta_J d)(c Theta^J b)	True
12	Hartree-Fock energy E_H + E_x = sum_s (G_s/2) D_s with G = (1 - F^T) g (random density matrix, random couplings)	True
13	uniform matter R = (rho_S + rho Gamma17[0])/16: only D_0 = rho_S^2, D_1 = rho^2; with g_0 = gS, g_1 = gV, g_7 = -gA, g_8 = -gP the exchange energy is that of stage 2 (section 8.7)	True
14	uniform density matrix: D_0 = rho_S^2, D_1 = rho^2, D_2..D_8 = 0	True
15	stage-2 couplings gS = -100000, gV = 10000: G_0 = -98750, G_1 = 20000 (the HF couplings of section 8.7)	True
16	nonlinear field equation: dL_int/d psibar_a = -sum_r g_r sum_K (psibar Theta_K psi)(Theta^K psi)_a	True
17	the chirality chi17 commutes with every generator Sigma^AB, hence with every spin connection Omega_mu	True
18	curved chiral equations: E_L^dag Gamma0 (i Gamma^mu D_mu - m) psi = (i d0 + i(3Ha/2 + 2Hb) + tauO.d/(a,b)) phi_L - m phi_R	True
19	curved chiral equations: E_R^dag Gamma0 (i Gamma^mu D_mu - m) psi = (i d0 + i(3Ha/2 + 2Hb) - tauO.d/(a,b)) phi_R - m phi_L	True
20	flat limit a = b = 1, kF = kF0 (stage-2 saturation): m* = 0.3480973401..., E/A - m = -0.0283214279...	True
21	flat limit: isotropic pressure pa = pb = rho eps_F - eps (Gibbs-Duhem, eps_F = V0 + sqrt(kF^2 + m*^2))	True
22	energy-momentum conservation eps' + 3Ha(eps + pa) + 4Hb(eps + pb) = 0 for the KS matter (arbitrary a, b; relative residual)	True
23	primordial: rho(x0) x0 = kF0^7/(105 pi^4) (density grows like 1/x0)	True
24	primordial: m* decreases toward 0 as x0 -> 0 (blue-shifted ordinary momenta, attractive scalar coupling)	True
25	modern: eps a^3 -> constant and pa/eps -> 0 (dirac16complex behaves as pressureless matter in ordinary space)	True
26	far future: rho = rho0 e^{-3 H x0} and 1 - m* -> 0 like rho	True
27	the 00 closed forms of E^(1), E^(2), E^(3) (pair-set algorithm, tools/lovelock71.py) agree with the closed forms of section 12.6 of 7+1.nb	True
28	Einstein-Lovelock equations are linear in (Ha', Hb') (quasi-linear second-order system)	True
29	the constraint Sum alpha_k E^(k)0_0 - Lambda = kappa eps is preserved by the evolution (relative violation < 1e-7) in both cases	True

10. How to reproduce

1. `python3 tools/make_7p1_nb.py` writes 7+1.nb (unevaluated).
2. Evaluate 7+1.nb in a fresh kernel of Wolfram 15.0.1 (Evaluation ▷ Quit Kernel, then Evaluation ▷ Evaluate Notebook; or `Get["D:/Developer/github/7+1/tools/run_eval.wl"]`) and save it; the last section summarizes all checks.
3. `python3 tools/ref_ks_gravity.py` repeats the checks of this document independently (exit code 0 if all pass).
4. `python3 tools/make_docs.py` regenerates this document, its symbol dictionary and its provenance record.